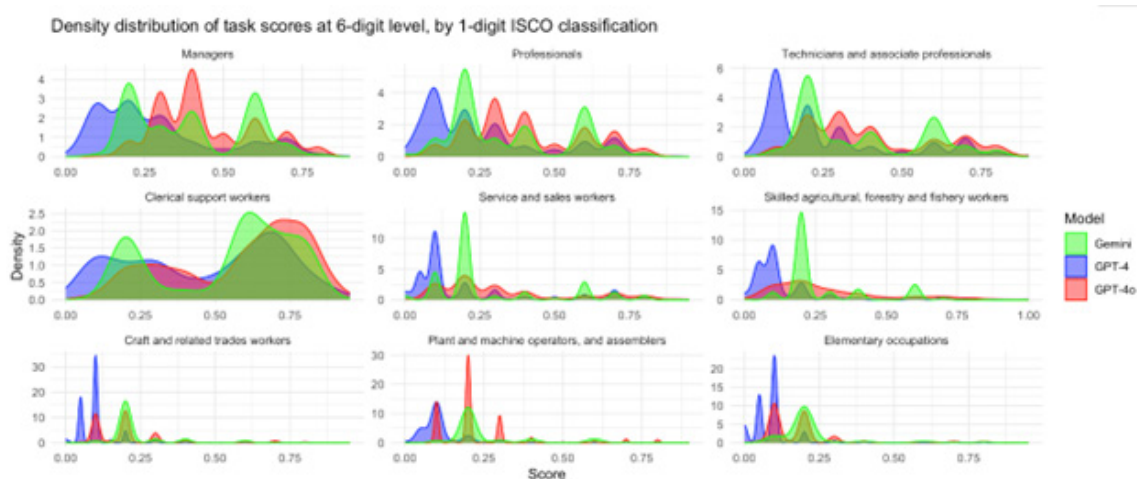


► 3 Assessment of tasks' automation potential

3.1. Initial Algorithmic predictions

As the first step, we conducted a purely algorithmic assessment of potential exposure to automation for each task in the Polish 6-digit system, relying on three advanced LLMs available on the market at the end of 2024: GPT-4, GPT-4o and Gemini Flash 1.5. The method relies on the approach previously used by GBB (2023), with a sequential API call devised to assign a score ranging from 0 to 1 for each individual task, where 0 means the task cannot be automated and 1 means the task could be fully automated with GenAI technology (the task can be conducted without participation of a human operator).⁴ The LLM is provided with information on the occupation to which the task pertains in ISCO-08 and requested to produce a justification for each predicted score.

► **Figure 2. Distribution of synthetic automation scores from 3 LLMs, by ISCO-08 1-digit**



The overall distribution of these synthetic automation scores is generally consistent with scores based on ISCO-08 documentation (GBB, 2023; Gmyrek, Winkler and Garganta, 2024) and O*NET (Elondou, 2023; Felten, 2023): the highest exposure to GenAI is typically observed among groups with the largest share of tasks of cognitive nature and it declines with tasks of physical nature, which are more prominent among occupational groups 4-9 (Figure 2). The largest density of high automation scores is observed among clerical workers and is stable across the three LLMs. In other groups, we observe some important differences in the LLMs' predictions. In particular, Gemini Flash 1.5 seems to provide a bimodal scoring, with most scores falling either below 0.25 or above 0.6. The other two models provide a wider range of scores, but still with important divergences, which makes it difficult to judge which model one should follow.⁵ Therefore, we do not examine these distributions in greater detail but proceed directly to improvements that could enhance coherence and stability of our predictions, regardless of the LLM applied.

⁴ In GBB (2023), the scores and justifications were assessed for consistency and stability of predictions over time, with the written justifications reviewed by humans. Tasks with scores above 0.8 (high possibility of automation) were transformed into embeddings, with a semantic clustering algorithm applied to identify the major groups of such tasks, which are subsequently reviewed by humans.

⁵ A comparison to GBB (2023) scores suggests the closest alignment with GPT-4o synthetic scoring of the Polish tasks' database.

3.2. Selection of occupational tasks for human assessment

To improve the precision of our synthetic predictions, we design a human-based evaluation of the potential for task automation. Since scoring of all 29,753 tasks would be too cumbersome for a survey, we focus on a sub-sample of tasks that are representative of the tasks in each professional group at 1-digit ISCO-08 level (Table 1). We employ the standard formula to calculate sample size (see Appendix 2).

Table 2 presents task sample calculations for each 1-digit category. The minimum number of scorings per question is set to 20, based on the general rule used by ILO Statistics for micro-data analysis, where observations with less than 16 respondents are flagged as unreliable. The number of scorings is set to 35 per respondent, based on our pre-testing of the survey form for potential fatigue effects.

► Table 2. Survey sample selection

ISCO-08 1-digit category	Description (1-digit)	Number of tasks	% of all tasks	Unadjusted Sample Size (Infinite Population)	Adjusted sample size	Sample Proportion	Nr of scorings per task	Total task scorings	Tasks per survey respondent	Nr of Respondents
1	Managers	2'565	8.6%	384	334	11.7%	20	6682	35	191
2	Professionals	8'715	28.6%	384	368	12.9%	20	7357	35	210
3	Technicians And Associate Professionals	5'956	21.1%	384	361	12.6%	20	7216	35	206
4	Clerical Support Workers	768	2.6%	384	256	9.0%	20	5124	35	146
5	Service And Sales Workers	1'730	5.4%	384	314	11.0%	20	6288	35	180
6	Skilled Agricultural, Forestry, And Fishery Workers	684	2.3%	384	246	8.6%	20	4923	35	141
7	Craft And Related Trades Workers	4'554	15.5%	384	354	12.4%	20	7084	35	202
8	Plant And Machine Operators, And Assemblers	3'769	12.6%	384	349	12.2%	20	6972	35	199
9	Elementary Occupations	1'012	3.3%	384	279	9.7%	20	5571	35	159
0	Armed Forces Occupations									
	TOTAL	297'53	100.0%		2861	100%		57218		1635

Within these initial parameters, each respondent is assigned a random set of tasks from the larger sample of tasks for each broad occupational group. The randomization of the tasks' sample selection maintains a general pattern of the distribution of scores similar to the scores initially generated with the LLMs (Section 4.1). However, the respondents are not made aware of these prior distributions, instead they receive a wide range of tasks with respect to their automation potential.

3.3. Survey design and recruitment

We conducted the human survey using the Computer-Assisted Web Interview (CAWI) method, drawing on a large online research platform in Poland, used for both commercial and scientific studies. This also means that some demographic characteristics are already pre-filtered at survey deployment stage, whereas others are captured through a set of core questions at the start of the survey (Table 3). Specifically, we capture respondents' gender, age group, size and place of residence, region of residence, highest level of education and their current profession. The professional classification is further facilitated by providing respondents with a short description of ISCO-08 1-digit classifications, alongside examples of occupations that fit into each category.

In addition to demographic information, subsequent core questions are designed to capture respondents' knowledge and experience with GenAI. These questions help contextualize how familiarity and interaction with GenAI may influence assessments of task automation potential. Specifically, respondents are asked how long they have worked in their current profession (Q4), as more experienced workers may have a deeper understanding of their tasks and offer more cautious or realistic assessments. Additionally, they are queried on their familiarity with AI (Q5), with more knowledgeable respondents potentially providing more informed evaluations. We also capture how often respondents use AI tools in their job (Q6), which provides insight into their practical engagement with AI and its integration into their daily tasks.

Finally, in core questions 7-8, we ask how respondents view the broader impact of AI on jobs in their field, helping us understand their perspectives on automation over the next five years. The final question in this section probes respondents' views on the likelihood that their own job could be partially or fully replaced by AI, which helps assess whether job security concerns might influence their task evaluations. Table 3 below provides a summary of the core questions.

► **Table 3. Core survey questions**

	Question	Answer Options
1	What is your gender?	Male / Female / Other / Prefer not to say
2	What is your age?	Age ranges (e.g., 18-24, 25-34, 35-44, etc.)
3	What is your highest level of education?	1. High school or equivalent; 2. Bachelor's degree; 3. Master's degree; 4. PhD or higher; 5. Other
4	How many years have you worked in your current profession?	1. Less than 1 year; 2. 1-5 years; 3. 6-10 years; 4. 11-15 years; 5. More than 15 years
5	Have you heard of Generative Artificial Intelligence (GenAI, e.g. ChatGPT)?	1. Yes; 2. No; 3. Hard to say
5a	How would you rate your level of knowledge about Generative Artificial Intelligence (GenAI, e.g. ChatGPT)?	1. Very low; 2. Low; 3. Average; 4. High 5. Very high
6	How often do you use Generative Artificial Intelligence tools (GenAI, e.g. ChatGPT)?	1. Never; 2. Rarely (a few times a year); 3. Occasionally (a few times a month); 4. Frequently (a few times a week); 5. Daily
7	How do you think Generative Artificial Intelligence (GenAI, e.g. ChatGPT) will impact jobs in your field in the next 5 years?	1. No impact at all; 2. Slight impact; 3. Moderate impact; 4. Strong impact; 5. Transformational impact